

Hardware/Software Co-Design for MAYO Verification

Final Presentation

Kowsyap Pranay Kumar Musumuri

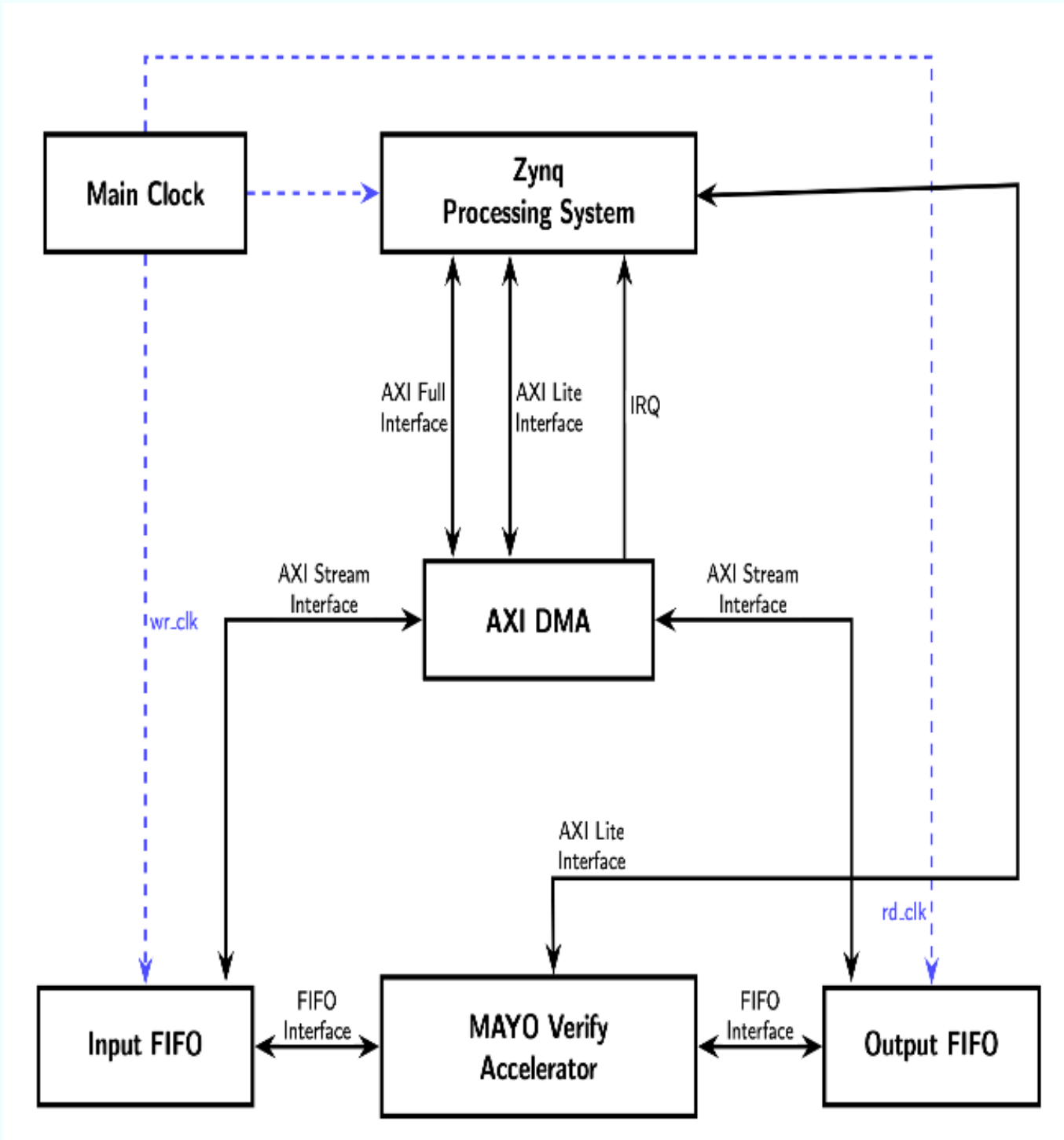
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ECE 615

System Architecture

- Host PC → UART → Zynq PS (ARM Cortex-A9, bare-metal C)
- PS: mayo_expand_pk() — CPK → EPK via AES-128-CTR
- PS: deriveT() — SHAKE256(hash(msg) || salt) → target vector T
- PS → AXI DMA → streams SIG || EPK to PL
- PL: sig_decoder, epk_decoder, quad_acc → Y
- PS: compare Y == T → send 0x01 (PASS) / 0x00 (FAIL) via UART

Parameter	Round 1 (R1)	Round 2 (R2)
m (equations / GF(16) outputs)	64	64
n (variables)	78	81
o (oil variables)	18	17
k (signature vectors)	4	4
cpk (bytes)	5,488 B	4,912 B
sig (bytes)	180 B	186 B
epk (bytes)	98,592 B	106,272 B
salt	24 B	24 B
digest	32 B	32 B
pk_seed	16 B	16 B



sig_decoder → s vectors
epk_decoder → P1, P2, P3 matrices
quad_accumulator → $Y = s^T \cdot P \cdot s$ in GF(16)

$$P_a = \begin{bmatrix} P1_a & P2_a \\ 0 & P3_a \end{bmatrix} \in GF(16)^{n \times n}$$

$P1_a \in GF(16)^{(n-o) \times (n-o)}$ upper triangular (vinegar variables)
 $P2_a \in GF(16)^{(n-o) \times o}$
 $P3_a \in GF(16)^{o \times o}$ upper triangular (o = oil variables)

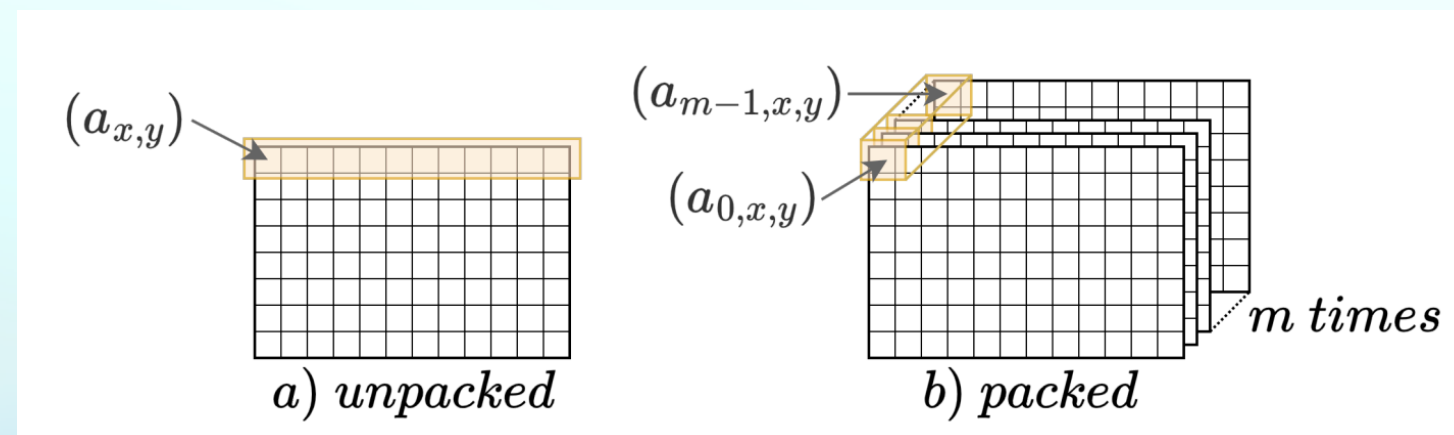
$y \in GF(16)^m$
 $P_a \in GF(16)^{n \times n}$
 $s_i \in GF(16)^{n \times 1}$

for i = 0 to k-1:
 for j = i to k-1:
 $y \leftarrow y + (s_i^T \cdot P_a \cdot s_j)$ for all a ∈ [m]

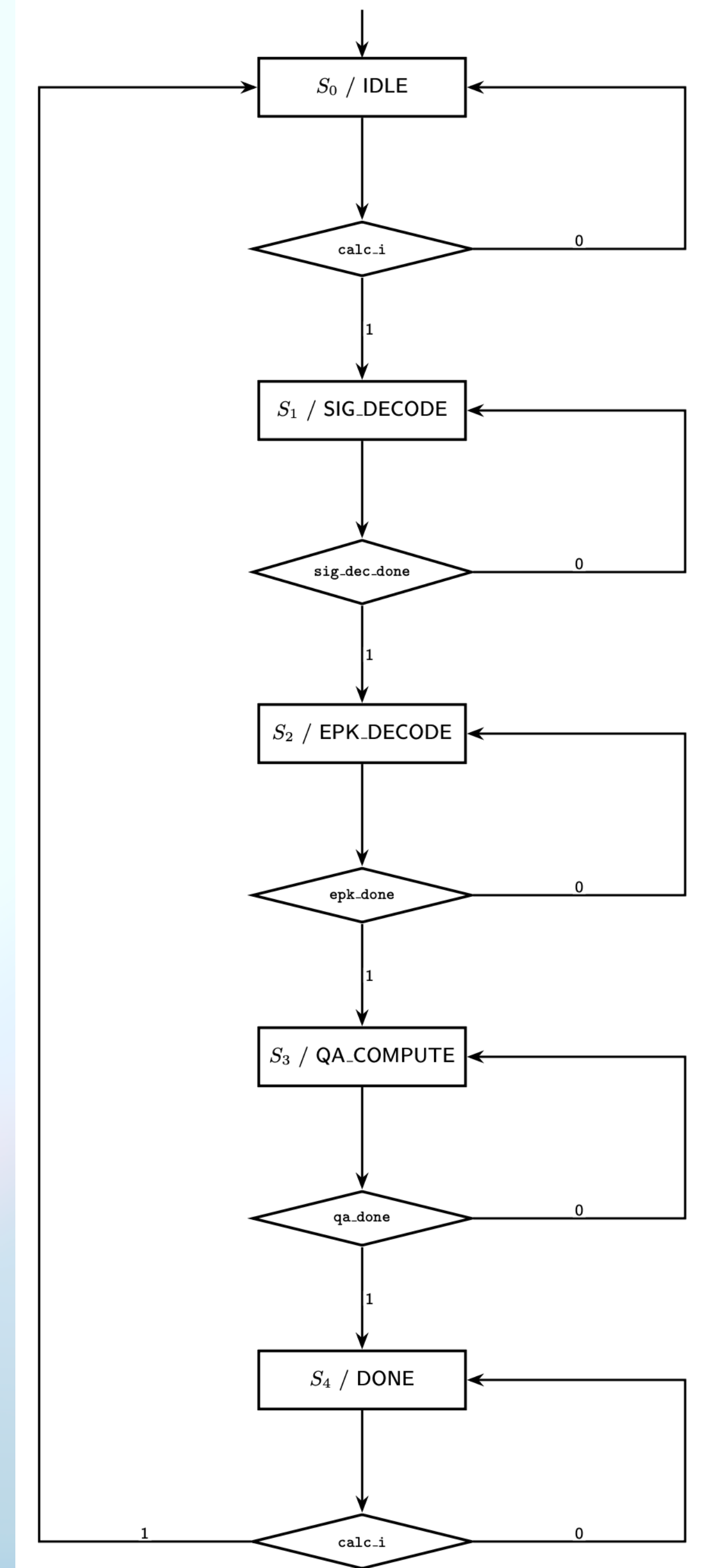
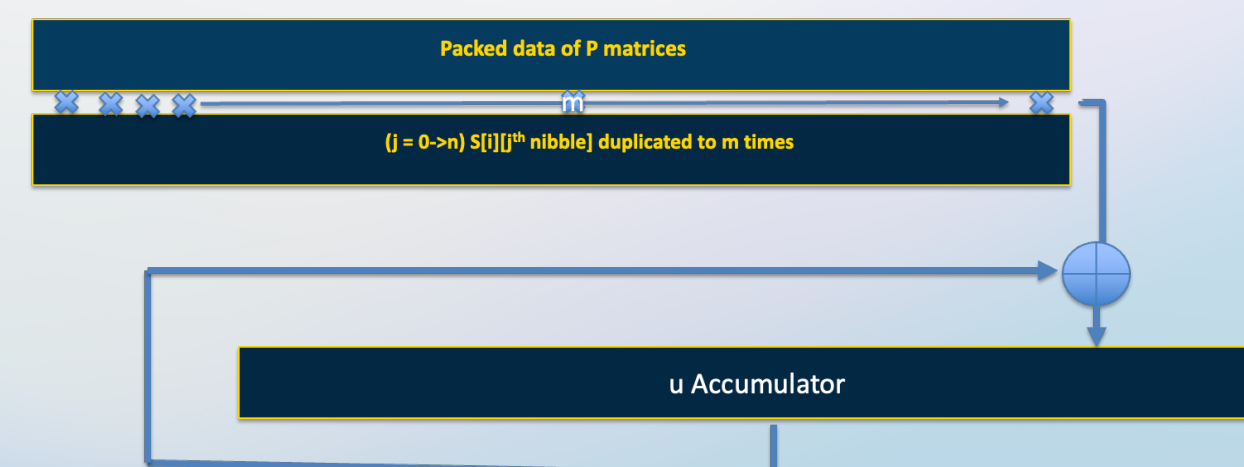
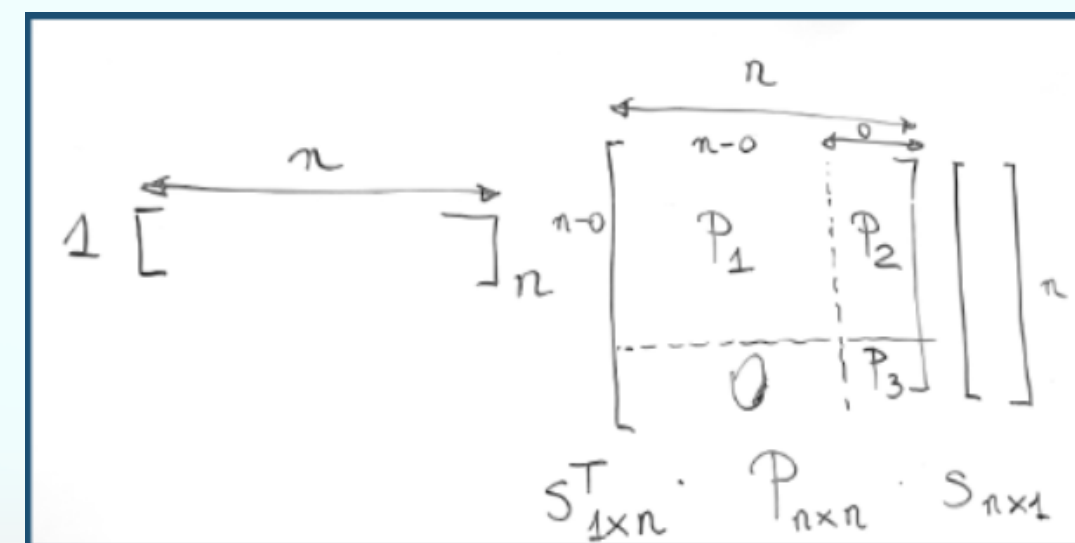
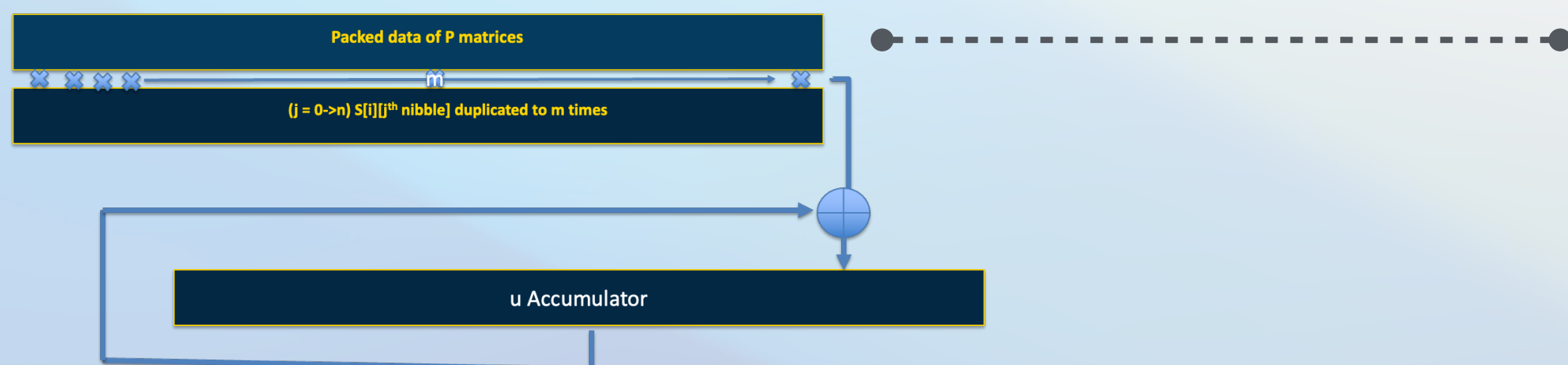
Y == T → SIGNATURE VALID / INVALID

Hardware Architecture

- AXI-Lite slave — bit 0: write 0x1 to start, read STATUS_DONE
- AXI Stream slave/master — receives SIG || EPK, emits Y result via DMA
- sig_decoder — unpacks k=4 compressed signature vectors (nibble alignment)
- epk_decoder — reads P_1, P_2, P_3 from stream into multi-bank SDP RAM
- quad_accumulator — computes $Y = s^T \cdot P \cdot s$ via parallel GF(16) multipliers



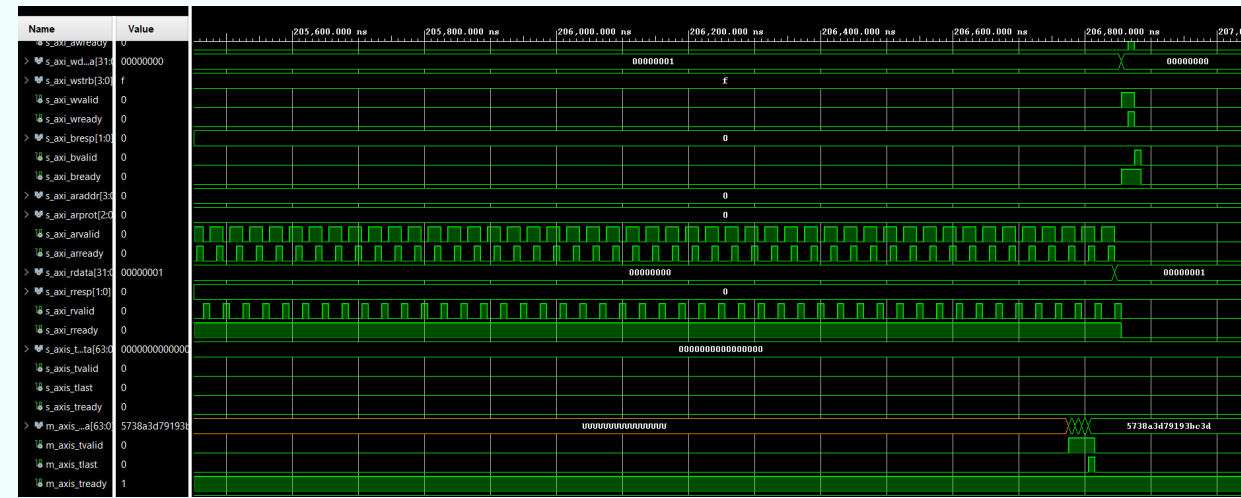
k times



Results

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 2.496 ns	Worst Hold Slack (WHS): 0.011 ns	Worst Pulse Width Slack (WPWS): 8.750 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 28900	Total Number of Endpoints: 28900	Total Number of Endpoints: 11200

All user specified timing constraints are met.



```
Note: calc asserted - starting combined SIG+RFX stream
Time: 145 ns Iteration: 0 Process: /mayo_verify_accelerator_tb/stimulus_proc File: C:/Users/Yakat/Downloads/code/hw_test/mayo_verify_acc
777822569b262AE
62193c2a65239220
8007551B4A8CF7E2
5738A3D79193BC3D LAST
Note: done observed through AXI-Lite status
Time: 206955 ns Iteration: 0 Process: /mayo_verify_accelerator_tb/stimulus_proc File: C:/Users/Yakat/Downloads/code/hw_test/mayo_verify_
Note: Simulation complete
Time: 206995 ns Iteration: 0 Process: /mayo_verify_accelerator_tb/stimulus_proc File: C:/Users/Yakat/Downloads/code/hw_test/mayo_verify_
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Resource	Round 1 (R1)	Round 2 (R2)
Slice LUTs	15,169 / 17,600 (86.2%)	15,748 / 17,600 (89.5%)
Slice Registers	10,795 / 35,200 (30.7%)	10,926 / 35,200 (31.0%)
Block RAM Tiles	23 / 60 (38.3%)	23 / 60 (38.3%)

- FPGA-based hardware accelerator for MAYO post-quantum signature verification
- PS/PL co-design on Zynq Z7-10 — ARM handles crypto expansion, PL evaluates GF(16)
- GF(16) quadratic evaluation fully parallelized → ~1.2 μ s hardware compute
- Supports both MAYO Round 1 and Round 2, Security Level 1 (Set 2)
- DMA transfer (1,116.7 ms) is current bottleneck — dominates total latency
- All Known Answer Tests (KAT) passed successfully, with both R1 and R2
- <https://github.com/kowsyap/mayo-verify-accelerator>

Operation	Time
mayo_expand_pk	73.9 ms
deriveT	0.17 ms
DMA TX (SIG + EPK)	1,116.7 ms
Hardware Compute	~1.2 μ s
Total	~1,191 ms